

Benjamin Chou

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Professional Summary

4th-year ECE PhD Student at Purdue University specializing in AI/ML research for audio, music, and multimodal systems. Researching generative model steering, stem separation, Transformers for error detection, and training-free strategies for Multimodal LLMs. I am proficient in Audio DSP, Python, PyTorch, TensorFlow, and HPC clusters. I also have organizational and communication skills (experience organizing 3 AI4Music workshops).

Education

Ph.D. in Electrical & Computer Engineering | *Purdue University*, West Lafayette, IN 2023 - Present

- GPA: 4.0/4.0

B.Sc. in Electrical Engineering | *National Cheng Kung University*, Tainan, Taiwan 2020 - 2023

- GPA: 4.0/4.3

Experience

Audio Applied Research Scientist Intern | *Shure*, Niles, IL Jul '26 – Sep '26

- Developing real-time speech enhancement and separation systems.

PhD Resident | *Google X*, Mountain View, CA Apr '26 – Jul '26

- Building a fully autonomous supply chain with LLM autonomous-loop orchestration and knowledge graphs for LLMs to coordinate planning, reasoning, and execution.
- Developing LLM agents for code review, deep research, and web surfing (LangChain, Vertex AI, RAG, coding agents, agent skills, loop engineering, Google Cloud Platform ([GCP](#)), GitHub CLI, Linear).

Applied Research Scientist Intern | *Reality Defender*, Manhattan, New York May '25 – Sep '25

- Developed an in-context learning framework with large audio language models, achieving SOTA deepfake detection on in-the-wild speech (+54.9% avg Macro F1 improvement) ([Natural Language Processing](#), [RAG](#), [ICL](#), [Audio LLMs](#), and [model routing](#)).

Graduate Research Assistant | *Under Dr. Yung-Hsiang Lu*, *Purdue University*, West Lafayette, IN Jan '24 – Present

- Developed an AI tutoring model with 6× higher F1 (AAAI 2025) and LadderSym with 2× better missed-note detection (ICLR 2026) ([PyTorch](#), [NLP](#), [Audio Transformers](#)).

- Leveraged HPC clusters and Linux environments to engineer generative pipelines for music mistakes, producing two datasets.

- Organized 'AI for Music' workshops at AAAI 2025, IEEE ICME 2025, and NeurIPS 2025 ([leadership](#), [communication](#)).

Research Assistant | *Under Dr. Hsun-Ping Hsieh*, *National Cheng Kung University*, Taiwan Mar '23 – Aug 2023

- Raised cancer-screening invitation accuracy to 83% using scikit-learn and tree-based classification ([machine learning](#), [statistics](#), [data science](#)).

- Elevated flash-chip yield by 17% with ensembling and Gaussian processes ([NumPy](#), [pandas](#)).

Projects

Improving Developer Code Understanding with GitHub Issues and Retrieval-Augmented Generation | [ChatGPT API](#), [REST](#), [Git](#)

- Built a RAG-based doc tool with GitHub issues; raised README clarity by 116% and code clarity by 16%.

Multi-Agent Self-Play for Beating Atari Games | [OpenAI Gym](#), [PettingZoo](#), [PyTorch](#), [RL](#), [Multi-agent systems](#)

- Used DQN/PPO self-play to improve Atari win rates by 11–32%.

Spectral Image Inpainting with Deep Learning | [MATLAB](#), [C++](#), [CUDA](#)

- Recovered 98% corrupted hyperspectral data at 0.02 RMSE using convex optimization and deep learning.

Honors & Awards

- Outstanding Student Scholarship, awarded to top 2 students in NCKU-Purdue program (2020–2023).

- AAAI Student Scholarship (2025), selected for NeurIPS volunteer (2025).

Publications

- **B. Chou**, Y. Zhu, S. Koppiseti, "[ICLAD: In-Context Learning with Comparison-Guidance for Audio Deepfake Detection](#)", *ACL 2026 Findings*

- **B. Chou**, P. Jajal, N. J. Eliopoulos, J. C. Davis, G. K. Thiravathukal, K. Y.-J. Yun, Y.-H. Lu, "[LadderSym: A Multimodal Interleaved Transformer for Music Practice Error Detection](#)", *ICLR 2026 (top 3.85%)*

- **B. Chou**, et al., "Enabling Note-Wise Music Separation and Editing via a Partially Synthetic Data Engine", *In Progress*

- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y. Lu, "[AdaPerceiver: Transformers with Adaptive Width, Depth, and Tokens](#)", *CVPR 2026 Findings*

- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y. Lu, "[Inference-Time Alignment of Diffusion Models with Evolutionary Algorithms](#)", *CVPR 2026 Findings*

- **B. Chou**, P. Jajal, N. J. Eliopoulos, T. Nadolsky, C.-Y. Yang, N. Ravi, J. C. Davis, K. Y.-J. Yun, Y. Lu, "[Detecting Performance Errors with Transformers](#)", *AAAI 2025*

- P. Jajal, N. J. Eliopoulos, **B. Chou**, G. K. Thiravathukal, J. C. Davis, Y.-H. Lu, "[Token Turing Machines are Efficient Vision Models](#)", *WACV 2025*

Patent Pending

- System and Method for Detecting Musical Performance Errors, Patent Application 2025.

- ICLAD: In-Context Learning with Comparison-Guidance for Audio Deepfake Detection, Provisional Patent Application 2025.